

Multi-Robot Coverage Path Planning using DARP on Crazyflie Drones

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Abstract—This paper presents the implementation of a coverage path planning algorithm combined with a divide-areas strategy for multi-UAV systems. The proposed approach is validated using Crazyflie UAVs operating in an indoor environment. The system is developed within the ROS 2 framework and relies on the Loco Positioning System for accurate indoor localization. The combination of area division and coverage path planning enables efficient space exploration while reducing overlap among multiple agents. Experimental results demonstrate the feasibility of the proposed solution for coordinated multi-UAV coverage tasks with the Crazyflies UAVs.

Index Terms—Crazyflie UAVs, Multi-Robot Systems, Coverage Path Planning, DARP, ROS2, Gazebo

I. INTRODUCTION

In recent years, the use of unmanned aerial vehicles (UAVs) has increased significantly in applications such as inspection, exploration, mapping, and search and rescue. In particular, multi-UAV systems offer clear advantages over single-agent solutions by reducing mission completion time, increasing system robustness, and improving coverage efficiency. However, coordination of multiple UAVs remains a challenging problem.

Within this context, the coverage path planning (CPP), which is a fundamental problem in mobile robotics, aims to generate trajectories that ensure complete coverage of a target area while minimizing redundancy and inefficiencies. This is done by determining a path that passes over all points in the area of interest while avoiding obstacles. When extended to multi-UAV scenarios, CPP requires additional mechanisms to distribute the workload among agents in an effective manner. This capability is essential for various applications, including search and rescue operations, precision agriculture, floor cleaning, and environmental monitoring.

This project addresses these challenges by integrating the Divide Areas Algorithm for Robotics Platforms (DARP) [1] where the area of interest is partitioned into subregions assigned to individual UAVs.

The practical validation of such algorithms on real robotic platforms is essential to assess their feasibility beyond simulation. Micro-UAV platforms such as the Crazyflie 2.1 [2]

provide an appropriate testbed for research in aerial robotics due to their low cost, small size, and compatibility with modern robotic frameworks. In this work, the Crazyflie platform is integrated with ROS2 [3], enabling modular development and scalable multi-agent control. Furthermore, accurate indoor localization is achieved through the Loco Positioning System, which provides reliable position estimates in GPS-denied environments.

This paper presents the implementation of a coverage path planning algorithm combined with an area division strategy for multi-UAV systems experimentally evaluated in Gazebo [4] for high-fidelity simulation and in an indoor environment equipped with the Loco Positioning System. The main objective is to demonstrate the feasibility of coordinated multi-UAV coverage missions using the Crazyflies and to analyze the system setup from a practical perspective, focusing on the challenges encountered during the implementation process and the solutions adopted to overcome them. This includes issues related to multi-UAV coordination, communication, localization, and integration within the ROS 2 framework, providing insights that may be valuable for future implementations on similar platforms.

The source code for this project is publicly available at <https://github.com/JoseLopez36/Crazyflie-RoboticsProjects>.

The remainder of this paper is organized as follows: Section II reviews the state of the art in Crazyflies. Section III details the system implementation and architecture. Section IV presents the simulation setup and results. Finally, Section V concludes the paper and discusses future work.

II. STATE OF THE ART

UAVs have been widely studied in recent years and continue to gain prominence in the research community because of their versatility and potential to be deployed in projects of a diverse nature, performing a wide range of tasks. This growing is partly driven by continuous advancements in robotics and micro-robotics, which have significantly lowered the entry barrier for research and experimentation. In particular, the emergence of low-cost platforms with high performance capabilities, such as the Crazyflie, has greatly improved acces-

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sibility to UAV technologies in both educational and research contexts.

A. Educational context

In the educational context, the Crazyflie has gained recognition as an accessible, easy-to-assemble platform for beginners, providing a first practical contact with the fields of robotics and UAVs. Its educational potential is increasingly highlighted in technical events, such as the 2024 Jornadas de Automática organised by the Comité Español de Automática, where Caballero-Martín et al. [5] showcased the advantages and practical benefits offered by the Crazyflie platform for learning and experimentation providing an overview of Crazyflie, its sensors and positioning systems, as well as the application of a PID controller.

In accordance with the educational objectives, Rafael Xavier Silva Cunha and Daniel Costa Ramos in their work [6] present an indoor experimental environment for UAV navigation to use in research and education. This integration utilizes the Loco Positioning System and allows students and researchers to conduct safe and scalable experiments in both simulated and physical settings.

B. Research context

Beyond its role in education, Crazyflie has been involved in various research contexts that are directly relevant to this work. Such as coverage planning.

For example, the work of Ana María Casado et al. [7] introduces a multi-UAV area coverage algorithm that leverages local inter-UAV interactions to achieve faster convergence. Using computational geometry techniques, the algorithm estimates uncovered regions and subsequently guides the UAVs toward those areas.

Along similar lines, the work of Alexandru Conțac and Adrian Burlacu [8] leverages Crazyflie micro-UAVs to develop an integrated framework for adaptive motion planning in complex indoor environments for small-scale aerial platforms.

May the work presented serve as sufficient and necessary justification for carrying out this work within the Crazyflie framework. Furthermore, we hope this work provide support or a baseline for future work based on the findings presented herein.

III. SYSTEM IMPLEMENTATION

The proposed system is developed within the ROS 2 (Robot Operating System 2) framework, leveraging its modularity and scalability for multi-robot applications. The architecture is designed to be agnostic to the underlying platform, supporting both high-fidelity simulation and real-world deployment on Crazyflie drones via the `crazyswarm2` interface.

A. System Architecture

The core logic is distributed across four custom ROS 2 nodes, each handling specific responsibilities in the coverage pipeline. The data flow and interaction between these components are described below.

1) *DARP Node*: The `darp_node` encapsulates the Divide Areas Algorithm for Robotics Platforms (DARP) logic. It functions as a service provider, exposing the `darp_service`. Upon receiving a mission request containing the operational area dimensions, obstacle positions, and initial robot positions, the node performs the following operations:

- **Discretization**: Converts continuous world coordinates into a discrete grid based on a configurable cell size.
- **Area Partitioning**: Executes the DARP algorithm to divide the grid into equal, connected, and non-overlapping zones for each agent.
- **Path Generation**: Computes a coverage path for each partition using a spanning tree algorithm.

The node returns the calculated trajectories (lists of waypoints) and a zone map, which encodes the area assignment for visualization, as illustrated in Fig. 1.

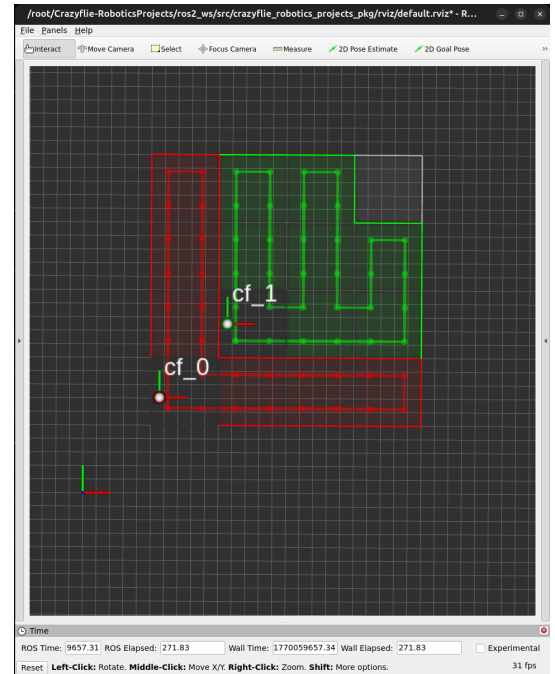


Fig. 1. DARP algorithm partitioning and trajectory generation in RViZ.

2) *Planning Node*: The `planning_node` acts as the central coordinator for the multi-robot team. Its primary roles include:

- **State Monitoring**: Subscribes to the real-time position of all agents via `/agent_id/state/position`.
- **Mission Orchestration**: Waits for all agents to initialize before requesting a plan from the `darp_node`. Once a solution is received, it distributes specific trajectories to each agent on their respective `/planning/trajectory` topics.
- **Dynamic Replanning**: Maintains a global map of visited cells. If a replan is triggered, previously visited areas are treated as virtual obstacles in the subsequent DARP request, ensuring efficient coverage without redundancy.

3) *Control Node*: A dedicated instance of the `control_node` runs for each UAV, managing its flight lifecycle through a finite state machine (FSM). The FSM states are:

- **INIT**: Performs system checks and sends initial setpoints.
- **TAKEOFF**: Commands the drone to ascend to a target altitude using the Takeoff service.
- **MISSION**: Executes the received trajectory. The node implements a path-following logic that switches to the next waypoint when the drone enters a defined acceptance radius of the current target.
- **LANDING**: Automatically initiates a landing maneuver upon trajectory completion.
- **LANDED**: Disarms the motors and terminates the operation.

This node interfaces with the lower-level flight controller via standard ROS 2 topics and services provided by the backend.

4) *Visualization Node*: The `visualization_node` is responsible for generating visual feedback in RViZ. It subscribes to agent positions, planned trajectories, and the zone grid, publishing corresponding `MarkerArray` messages. This allows for real-time monitoring of the coverage progress, grid partitioning, and flight paths.

Fig. 2 displays the architecture described.

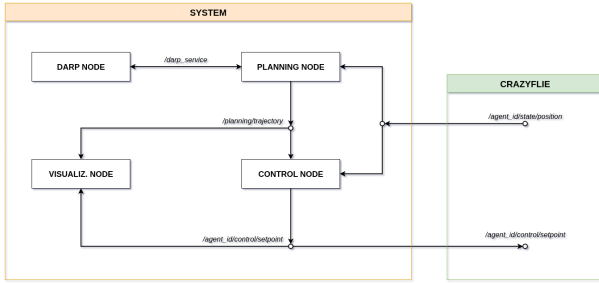


Fig. 2. System architecture implemented in ROS 2.

B. Simulation Setup

The system was developed and validated in a Docker-based environment to ensure reproducibility. The simulation relies on Gazebo paired with the `crazyswarm2` software-in-the-loop (SITL) backend. This setup accurately models the Crazyflie dynamics and allows for testing complex multi-agent scenarios before real-world deployment. The simulation setup is depicted in Fig. 3.

C. Real-World Implementation

The experimental validation was conducted using real hardware to assess the system's performance in a physical environment. The setup consists of the following components:

1) *Robotic Platform*: The experiments utilize Crazyflie 2.1¹ nano-quadcopters. Each drone is equipped with:

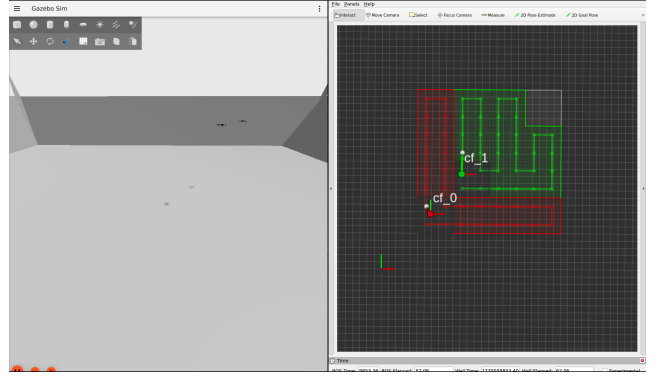


Fig. 3. Simulation environment in Gazebo with Crazyflie drones.

- **Loco Positioning Deck**²: configured in TDoA2 (Time Difference of Arrival 2) mode. TDoA2 is selected over TWR (Two-Way Ranging) as it supports an unlimited number of listeners, making it scalable for multi-robot fleets.
- **Flow Deck v2**³: provides optical flow and a ToF (Time-of-Flight) distance sensor. While the Loco Positioning System provides 3D coordinates, the vertical (Z-axis) estimation can be noisy. The Flow Deck is used to fuse stable height measurements, ensuring consistent altitude hold during the mission.

Communication between the ground station and the swarm is handled by a Crazyradio 2.0⁴ USB dongle.

2) *Localization System*: An indoor localization infrastructure, the Loco Positioning System (LPS), was deployed using 8 Loco Positioning Nodes⁵. To maximize the positioning volume and stability, the anchors were arranged in a cubic configuration encompassing the flight area. The nodes were placed at the following coordinates to create a diversified geometry:

- **Bottom Plane ($z \approx 0.75m$)**: Anchors at corners $(0, 0)$, $(3.6, 0.0)$, $(0, 3.6)$, and $(3.6, 3.6)$.
- **Top Plane ($z \approx 2.25m$)**: Anchors vertically offset from the bottom plane to ensure good geometric distribution.

This setup, shown in Fig. 4, provides accurate position estimates required for the precise grid-based navigation demanded by the DARP algorithm.

IV. RESULTS AND DISCUSSION

The transition from simulation to real hardware revealed significant practical challenges, mainly regarding localization precision and hardware reliability. The experimental validation was performed using two Crazyflie 2.1 drones in a 3.6×3.6 m flight area, with the Loco Positioning System (LPS) configured in TDoA2 mode to support multi-agent scalability as detailed in section III.

²<https://www.bitcraze.io/products/loco-positioning-deck/>

³<https://store.bitcraze.io/products/flow-deck-v2>

⁴<https://www.bitcraze.io/products/crazyradio-2-0/>

⁵<https://www.bitcraze.io/products/loco-positioning-node/>

¹<https://www.bitcraze.io/products/old-products/crazyflie-2-1/>



Fig. 4. Experimental setup with Crazyflie 2.1 and Loco Positioning System anchors.

To evaluate system performances, two experimental runs were chosen as representative case: a nominal run, in which the mission was completed successfully, and a degraded run, where the task could not be fully accomplished. This comparison helps identify the practical operational limits of the proposed approach.

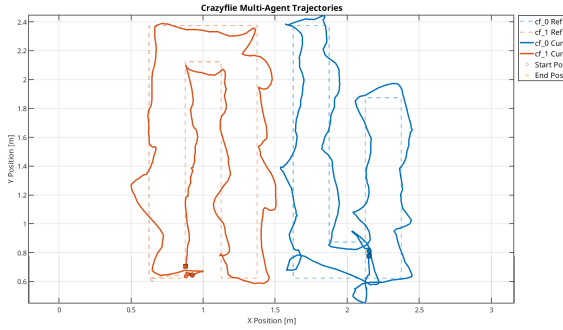


Fig. 5. Experimental results: nominal trajectory tracking performance.

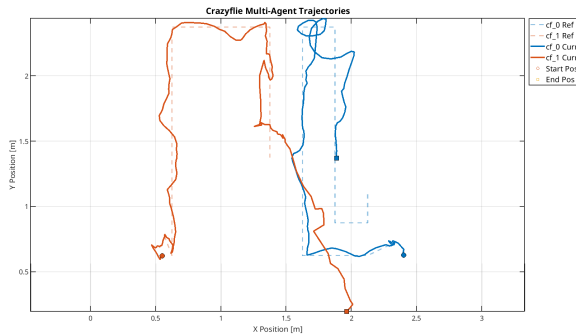


Fig. 6. Experimental results: degraded trajectory tracking performance.

1) *Nominal Mission Analysis:* As illustrated in Fig. 5, both agents were able to cover their assigned DARP regions, although noticeable deviations from the reference trajectories were observed. These deviations are mainly attributed to measurement noise introduced by the UWB-based localization

system. The quantitative performance indicators are summarized in Tab. I.

TABLE I
NOMINAL FLIGHT PERFORMANCE METRICS

Robot	Duration [s]	Distance [m]	Avg Speed [m/s]	RMSE [m]
cf_0	35.99	12.85	0.36	1.0317
cf_1	35.98	10.33	0.29	1.1950

Although RMSE values were relatively high, the system remained stable and completed the coverage task, showing tolerance to localization uncertainty.

2) *Degraded Mission Analysis:* In contrast, the degraded experiment shows unstable trajectories and incomplete coverage of the assigned regions. As shown in Tab II, both platforms experienced an increase in tracking error together with shorter flight durations, indicating premature termination of the mission.

TABLE II
DEGRADED FLIGHT PERFORMANCE METRICS (FAILURE CASE)

Robot	Duration [s]	Distance [m]	Avg Speed [m/s]	RMSE [m]
cf_0	20.79	6.23	0.30	1.2577
cf_1	22.48	8.38	0.37	1.3239

This performance degradation exposes a critical operating limit: as battery voltage decreases, the controller's capacity to compensate for TDoA2-related noise and communication delays is significantly reduced. The associated drop in thrust-to-weight ratio destabilizes state estimation and produces trajectory deviations that exceed the tolerance of the DARP grid, ultimately causing mission failure.

3) *Final Discussion:* The discrepancy between the flights, both using the TDoA2 protocol, allows for a deep technical critique of the system's limitations:

- **Battery-induced instability:** The failure in flight 2 was primarily caused by battery depletion after consecutive flights. As the voltage decreases, the Crazyflie's thrust-to-weight ratio decreases, making the drone unable to compensate for the high-frequency noise inherent in LPS.
- **Localization accuracy limits of TDoA2:** Both experiments consistently exhibited RMSE values above 1 m. This confirms that while TDoA2 allows for an unlimited number of agents, it introduces a noise floor that is often of the same order of magnitude as the acceptance radius used for waypoint switching. This results in the "oscillating" behavior observed in the plots as drones struggle to reach the precision required.
- **Multi-agent communication and interference:** We observed that precision is lower when two drones fly simultaneously compared to single-agent tests. Beyond UWB congestion, a significant factor is the use of a single Crazyradio 2.0 dongle to manage both agents on the same frequency channel. This setup forces the radio to multiplex communication across different addresses,

leading to increased latency and a higher rate of packet loss, contributing to the higher *RMSE* observed in multi-agent tests.

- **Importance of Flow Deck v2:** The Flow Deck v2 played a vital role in maintaining flight stability. In degraded battery states, the fusion of optical flow and ToF (Time-of-Flight) altitude data was the only factor preventing immediate ground collisions due to Z-axis noise from the LPS.

V. CONCLUSION

This paper presented the successful implementation of a multi-UAV coverage path planning framework using the DARP algorithm and ROS 2. The proposed system was validated through both high-fidelity simulations and real-world experiments in an indoor environment using Crazyflie 2.1 drones and the Loco Positioning System. The experimental results demonstrate that the combination of area division and grid-based path planning effectively ensures complete coverage while preventing agent overlap. However, the transition from simulation to physical hardware revealed a significant trade-off between scalability and localization precision:

- **System feasibility:** The architecture proved robust enough to maintain coordinated flight behavior even with a localization noise floor exceeding 1 meter *RMSE* in TDoA2 mode.
- **Hardware constraints:** We identified battery depletion and the resulting voltage drops as the primary causes of mission failure. These factors compromised the flight controller's stability, leading to the erratic trajectories and premature mission termination observed in degraded trials.
- **Communication and stability:** The use of a single Crazyradio 2.0 for multiple agents introduced latency that, coupled with the inherent noise of the LPS, challenged the stability of the EKF. Furthermore, the Flow Deck v2 was found to be indispensable for maintaining altitude hold in these power-constrained scenarios.

ACKNOWLEDGMENT

The authors would like to thank the teaching staff of the Robotics Projects course for their guidance and support.

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